huge number of people suddenly individually racking their brains to come up with something new could be exactly what we need. “Out of that explosion someone might have an idea. ‘Oh! This is the way to figure out the future.’”

How long such a program will take is another question. If we’re just trying to distinguish between a cosmological constant and some other form of dark energy, we literally have all the time in the world, and then some. There’s really no theory out there in which dark energy can destroy our planet before our own Sun does the job.

But vacuum decay is another matter. The Standard Model of particle physics, the very same one that has passed every experimental test we’ve come up with, places us in a precarious position on the edge of total universal instability. How likely this is to be an actual risk, or a quirk of the extrapolation of an incomplete theory, depends on who you ask. (For the record, I asked several experts and I got answers ranging from “it tells us our theory is wrong” to “the risk is really tiny” to “maybe we’ve just been lucky so far.” Take that as you will.) In any case, if we want to be able to say something more reassuring than “It’s useless to worry because you are not going to feel any pain,”[fn4] we’re going to need a very specific kind of data.

Fortunately, we have a pretty good idea where we can get it.

DISCOVERY MACHINES

There is no place on Earth with a more persistent, if wholly undeserved, association with the destruction of the cosmos than CERN. Best known as the home of the Large Hadron Collider, CERN is a sprawling campus of laboratories and office buildings covering about six square kilometers that straddles the France-Switzerland border near Geneva. It’s essentially an oddly specialized little border town, complete with its own fire department and post office, alongside laboratories and machine shops and a bona fide antimatter factory. Physicists at CERN have been accelerating and smashing protons since the 1950s, long before the LHC was built, carrying out increasingly complex and sensitive experiments to examine the nature of subatomic particles by obliterating them against each other. These kinds of experiments helped us to create the Standard Model of particle physics,